

A General Algorithm

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$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

$$\left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 8 \end{array} \right]$$

What are the similarities between these two augmented matrices?

Row Echelon Form

Definition

Row Echelon Form. We say that a matrix is in Row Echelon Form if and only if it satisfies the following requirements

1. All non-zero rows are above any rows of all zeros.
2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
3. All entries in a column below a leading entry are zeros.

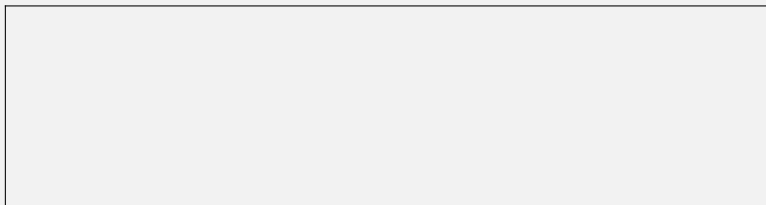
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Example



Reduced Row Echelon Form

Definition

Reduced Row Echelon Form (RREF). We say that a matrix is in RREF if and only if it satisfies the requirements for Row Echelon Form and

4. The leading entry in each non-zero row is 1.
5. Each leading 1 is the only non-zero in its column.

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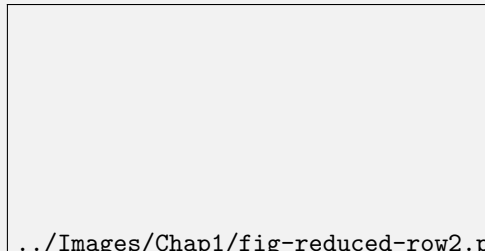
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../Images/Chap1/fig-row-echelon2.png



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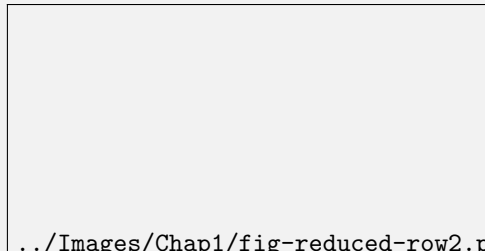
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Example



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Formalizing Our Algorithm

We've been doing this already actually! In order to better formalize it, let's break down our algorithm into 2 smaller steps.

Algorithm 1 RREF Transformation

Input: A matrix A with m rows and n columns of real numbers

Output: A transformed to RREF using Only Elementary Row Operations

- 1: Reduce A to Row Echelon Form
 - 2: Reduce A to RREF
-

Reducing to Row Echelon Form

Let's start with an example. Let's reduce the following augmented matrix to Row Echelon Form.

$$\left[\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 2 & 8 & 1 & 2 \\ 1 & 4 & 1 & 1 \end{array} \right]$$

1. Start with the leftmost non-zero column.
We call this the **pivot column**.

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1. Start with the leftmost non-zero column.
We call this the **pivot column**.
2. Pick some non-zero element. This is called the **pivot**. Move it to the top if necessary.

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 2 & 8 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{array} \right]$$

Reducing to Row Echelon Form

Let's start with an example. Let's reduce the following augmented matrix to Row Echelon Form.

1. Start with the leftmost non-zero column.
We call this the **pivot column**.
2. Pick some non-zero element. This is called the **pivot**. Move it to the top if necessary.
3. Create zeros in all positions below the pivot.

$$\left[\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 2 & 8 & 1 & 2 \\ 1 & 4 & 1 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 2 & 8 & 1 & 2 \\ 0 & 1 & 2 & 3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \end{array} \right]$$

Reducing to Row Echelon Form

4. Ignore the row containing the pivot and consider the remaining submatrix (highlighted in red). Then repeat the steps from before again!

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \end{array} \right]$$

Reducing to Row Echelon Form

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$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 2 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & -1 & 0 \end{array} \right]$$

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1. All our pivots are 1
2. All elements above our pivots are 0.

So, let's fix them!

Reducing to RREF

5. Scale each row to make all pivots equal to 1.

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Reducing to RREF

5. Scale each row to make all pivots equal to 1.
6. Starting with the rightmost pivot column, zero out all elements above each non-zero pivot using Elementary Row Operations.

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -11 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

Then our answer is

$$(x_1, x_2, x_3) = (-11, 3, 0)$$

Gaussian Elimination¹

Algorithm 2 Gaussian Elimination

Input: A matrix A with m rows and n columns of real numbers

Output: A transformed to RREF

- 1: Consider the leftmost column. Called the **pivot column**
 - 2: Pick a non-zero entry as the **pivot**. Exchange rows if needed.
 - 3: Use Elementary Row Operations to create zeros below the pivot
 - 4: Ignore the row containing the pivot. Repeat the 3 steps above on the remaining submatrix until there are no more non-zero rows.
 - 5: Scale each row so each non-zero pivot is 1.
 - 6: Cancel out all elements above each pivot with Elementary Row Operations in each pivot column.
-

¹https://en.wikipedia.org/wiki/Gaussian_elimination#History

Existence and Uniqueness of RREF

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$$\left[\begin{array}{cc|c} 0 & 1 & -2 \\ -2 & -2 & -4 \\ -1 & 0 & -4 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} -2 & -2 & -4 \\ 0 & 1 & -2 \\ -1 & 0 & -4 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & -2 \\ -1 & 0 & -4 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array} \right]$$

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Existence and Uniqueness of RREF

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Theorem

Every matrix is equivalent to one and only one reduced RREF matrix.

Example

Find a solution to the system

$$x_1 + 2x_2 + x_3 = -2$$

$$x_1 + 3x_2 - 2x_3 = 1$$

Using Gaussian Elimination, we get

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & -2 \\ 1 & 3 & -2 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & -2 \\ 0 & 1 & -3 & 3 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 7 & -8 \\ 0 & 1 & -3 & 3 \end{array} \right]$$

Example

Find a solution to the system

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This gives the solution set:

$$x_1 + 7x_3 = -8$$

$$x_2 - 3x_3 = 3$$

or in parametric form:

$$(-8 - 7x_3, 3 + 3x_3, x_3) = \begin{cases} x_1 = -8 - 7x_3 \\ x_2 = 3 + 3x_3 \\ x_3 \text{ is free} \end{cases}$$

Parametric Form

Definition

Parametric Form: We say that a solution to a system of linear equations is in **parametric form** if it has a variable in the ordered pair.

Example

For example

$$(-8 - 7x_3, 3 + 3x_3, x_3)$$

is in parametric form because x_3 is still a variable and we don't know what it is! We will usually write either the variable with a subscript like above or using another variable. Like:

$$(-8 - 7s, 3 + 3s, s)$$

Solutions of Linear Systems

We have

$$\begin{cases} x_1 = -8 - 7x_3 \\ x_2 = 3 + 3x_3 \\ x_3 \text{ is free} \end{cases}.$$

- ▶ Variables x_1 and x_2 came from **pivot columns**, and we call these **basic variables**
- ▶ Variable x_3 did not come from a pivot column, so we call it a **free variable**. This means it take on any value we want (IE we are free to choose it)
- ▶ If we have at least one free variable, then we have an infinite number of solutions.

Another Example

Find a solution to the system

$$\begin{aligned}x_2 + 2x_3 &= 0 \\x_1 + 2x_2 + 3x_3 &= 1 \\x_1 + x_2 + x_3 &= 2\end{aligned}$$

Gaussian Elimination gives us

$$\left[\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & 0 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

This is an inconsistent system!

$$\begin{aligned}x_1 - x_3 &= 0 \\x_2 + 2x_3 &= 0 \\0 &= 1\end{aligned}$$

Summary

For a given system of linear equations:

1. Write the augmented matrix
2. Use Elementary Row Operations to reduce to Row Echelon Form
 - 2.1 If the rightmost column is a pivot column, then it is inconsistent, and no solution exists. We can now stop
 - 2.2 If the rightmost column is not a pivot column, then it is consistent, and we have a solution, so we need to continue
3. Reduce to RREF
4. Rewrite as a system of linear equations
5. Put all basic variables on the left and all free variables on the right.

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